

Tyler Jay Yang

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EDUCATION

Carnegie Mellon University, School of Computer Science

Bachelor of Science (B.S.) in Computer Science, Concentration in Computer Systems

Pittsburgh, PA

Expected May 2028

Teaching Assistant, Parallel and Sequential Data Structures and Algorithms (15-210)

Spring 2026 - Present

Teaching Assistant, Multivariable Calculus (21-259)

Fall 2025

EXPERIENCE

Aven Financial

Software Engineer

May 2026 - Aug 2026

Polars | PostgreSQL | Snowflake | Prefect | Superset | Vue.js

- Saved **\$1.3M annually (\$25,000/week)** in growth spend by improving statistical precision of Direct Mail prescreening
- Screened **120M users/week** after overhauling existing infrastructure – building a recommender system for product eligibility screening and mail envelope/letter assignment via combinatorial optimization
- Cut cloud spending by **94%** by decreasing memory footprint (**32x lower RSS**) for backtests on the new infrastructure
- Generated **27 leads in 40 visits** by deploying regulation-compliant landing pages for Direct Mail recipients

Scale AI

Technical Advisor

Aug 2025 - Dec 2025

Python | C++ | Docker

- Staged multi-turn scenarios to evaluate model reasoning capabilities and agentic tool use
- Conducted 3 experiments to evaluate frontier AI dual-use capabilities in cybersecurity (e.g., web scraping, DDoS)

PROJECTS

C0 Compiler

Compiler Design

Jan 2026 - May 2026

OCaml | C | LLVM | x86-64 ASM | GDB | Valgrind

- **Wrote a compiler from scratch** for a well-defined subset of C, supporting dynamic memory allocation and multi-target compilation (assembly and LLVM IR)
- Achieved **up to 2x speedup over GCC -O1** on 19 core benchmarks by implementing multi-pass optimizations, including Sparse Conditional Constant Propagation, Aggressive Dead Code Elimination, Global Constant Subexpression Elimination

Optimizing CUDA Kernels and Schedulers for Tensor Graphs

Machine Learning Systems

Mar 2026 - Apr 2026

CUDA | TIRx DSL | C++

- Achieved **1,322 TFLOP/s (4.4x baseline)** of throughput on NVIDIA Blackwell (SM100) GPUs by implementing GEMM kernel using asynchronous memory load, warp specialization, 4-stage software pipelining, CTA cluster multi-casting
- Decreased tensor graph execution latency (**47% improvement over baseline**) measured on analytical roofline model by routing subgraphs to size-aware sub-solvers using dynamic programming and greedy chain-fusion heuristics

Multimodal Framework for Federated Learning (MuFFLe)

At CMU x NVIDIA Federated Learning Hackathon

Jan 2026

PyTorch | NVFlare | S3

- Won **Most Collaborative Award** from NVIDIA and AWS judges for prototyping extensible infrastructure to horizontally harmonize sensitive data across biobanks and vertically aggregate different data modalities for training deep learning models without leaking private health data. Preprint: https://osf.io/preprints/biohackrxiv/5psfj_v1

Backend Matchmaking Server

For ACM at CMU, Algorithms with a Purpose (AWAP) Hackathon

Dec 2025 - Jan 2026

Golang | Docker | Redis | S3 | EC2 | DynamoDB

- Ensured **zero memory leakage** by coordinating team of 5 engineers to refactor the matchmaking server to use a semaphore-throttled worker pool
- Co-organized algorithms hackathon for **200+** participants

SKILLS

Languages: Python, C, C++, Go, OCaml, SML, Zig, TypeScript, Java, Julia, R, MATLAB, Shell/Bash

Machine Learning & Data: PyTorch, TensorFlow, Scikit-Learn, NumPy, Polars, Pandas, WandB, Ray, Matplotlib

Data Infrastructure & Databases: PostgreSQL, Snowflake, Redis, Superset, Prefect, SLURM

Cloud & DevOps: AWS (EC2, S3, DynamoDB), Docker, Git

Developer Tools: Vim, Jupyter, Linux/Unix